NUMBER OF POINTS OF VARIETIES IN FINITE FIELDS.*

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1. Statement and proof of the main theorem. Let V be a variety in a projective space P^n. If V has dimension r and degree d, we shall indicate this by writing $V = V_{n,d,r}$. A point of V can be represented by homogeneous coordinates $(x) = (x_0, \cdots, x_n)$, and we shall also say that (x) is a point of V. We use k to denote a finite field and q to denote the number of elements in k. Let V be defined over k. A point (x) of V is said to be rational over k, or more briefly to be in k, if its coordinate ratios x_i/x_j $(x_j \neq 0)$ are in k. We intend to give an estimate for the number of points of V which are in k. Denoting this number by N, we prove

THEOREM 1. *There exists a constant $A(n, d, r)$ depending only on n, d, r such that for any variety $V = V_{n,d,r}$ defined over a finite field k we have*

$$| N - q^r | \leqq \delta q^{r-\frac{1}{2}} + A(n, d, r) q^{r-1}$$

where $\delta = (d-1)(d-2)$.

If $r = 1$, i. e. if V is a curve, the above diophantine statement is a reformulation of the Riemann hypothesis in function fields [2]. Indeed, let V_1 be a non singular projective curve, defined over k, birationally equivalent to V over k. Let g be the genus of V and V_1. Then we know that the number N_1 of rational points of V_1 satisfies $| 1 + q - N_1 | \leqq 2gq^{\frac{1}{2}}$. Each non singular point of V corresponds to exactly one point of V_1. The number of singular points on V is bounded by a constant depending only on d, and each singular point of V corresponds at most to d points on V_1. Hence $| N_1 - N | \leqq A$ where A is a constant depending only on d. This shows that $| N - q | \leqq 2gq^{\frac{1}{2}} + (A + 1)$. Using the fact that $g \leqq \frac{1}{2}(d-1)(d-2)$, we see that Theorem 1 is true for $r = 1$.

The proof of Theorem 1 will now be carried out by induction on the dimension r, and the arguments will be of an elementary nature.

We begin by two lemmas and use throughout the terminology of [1],

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except that a bunch of varieties will be called an algebraic set. We shall say that an algebraic set is defined over k if it is defined by algebraic equations $f_\alpha = 0$ with coefficients in k. If the algebraic set is in projective space, then the f_α can be chosen to be forms, and the algebraic set is identified with the rays of zeros of these forms. For example, a hyperplane H in P^n is defined by an equation $\sum w_i X_i = 0$. If $w_i \in k$, then H is defined over k, or as we shall say more briefly, H is in k.

Let Z be a positive cycle in P^n, of degree $d > 0$, and dimension r. We can then write $Z = \sum a_i V_i$ as a formal sum of distinct varieties V_i, with integer coefficients $a_i > 0$. We then have $d = \deg Z = \sum a_i \cdot \deg V_i$, and $\dim V_i = r$ for each i. By a point of Z we shall mean a point of any of the varieties V_i. The set of points of Z is an algebraic set denoted by $|Z|$. If Z is rational over k, then $|Z|$ is defined over k. We let N_Z be the number of points of Z in k.

LEMMA 1. *There exists a constant $A_1(n, d, r)$ depending only on n, d, r such that for any positive cycle Z in P^n, of degree d, dimension r, and rational over k, we have $N_Z \leqq A_1 q^r$.*

Proof. By induction on the dimension r. Suppose first $r = 0$. Then $|Z|$ consists of at most d points, and the lemma is trivial. Assume now $r \geqq 1$. If we express Z as a sum of prime rational cycles over k, then there will be at most d such cycles in the sum. Hence it suffices to prove the result for a prime rational cycle. Let $(X_0, \cdots, X_n)$ be the variables of P^n. For some pair of indices, say $(0, 1)$, Z (which we assume to be prime rational over k) intersects properly the hyperplanes H_ξ defined by the equations $X_0 - \xi X_1 = 0$ where ξ ranges over k, and the hyperplane H defined by $X_1 = 0$. Every point of Z in k is contained in one of the cycles $Z \cdot H_\xi$ or $Z \cdot H$, each of which has degree d by Bezout's Theorem ([1], App. I) and is of dimension $r - 1$. According to the induction hypothesis, there exists a constant $B(n, d, r-1)$ such that $N_{Z \cdot H_\xi}$ and $N_{Z \cdot H} \leqq B(n, d, r-1) q^{r-1}$. As we have exactly $q + 1$ cycles, we see that there are at most $B(q+1) q^{r-1}$ points in Z, and this is certainly $\leqq 2Bq^r$. The constant $2B$ is what we are looking for.

Our second lemma will be concerned with the following situation. Let V be a variety in P^n, of dimension $r \geqq 2$, and not contained in a hyperplane. Let P'^n be the projective space dual to P^n, and let $(w) = (w_0, \cdots, w_n)$ be the homogeneous coordinates of a point in P'^n. Let H_w be the hyperplane defined by $\sum w_i X_i = 0$. Then the set R of points (w) of P' such that the cycle $V \cdot H_w$ is not a variety is an algebraic set, defined over every field of definition of V, and $R \neq P'$. (Cf. [4], § 1, 6). The hyperplanes H such that $V \cdot H$ is not a variety may therefore be viewed as forming an algebraic set R in P'. Lemma 2 gives an estimate for the number of such hyperplanes defined over a finite field k.

LEMMA 2. *There exists a constant $A_2(n, d, r)$ depending only on n, d, r having the following property. If $V = V_{n,d,r}$ is any variety defined over k, not contained in a hyperplane, and if N_R is the number of hyperplanes H in k such that $V \cdot H$ is not a variety, then $N_R \leqq A_2 q^{n-1}$.*

Proof. Let Z be a positive cycle of dimension r and degree d in P^n. Let $F(U^{(0)}, \cdots, U^{(r)})$ be the associated form of Z. It is of degree d in each set of variables $U^{(0)}, \cdots, U^{(r)}$, and its coefficients are called the Chow coordinates of Z. These are viewed as the homogeneous coordinates of a point in a projective space P^M. The coordinates $(c) = (c_0, \cdots, c_M)$ of a cycle which is not a variety form an algebraic set C in P^M (cf. [4], § 1, 6), and C is defined over the prime field. We let $\phi_\alpha = 0$ be a finite system of equations with coefficients in the prime field, defining C. The forms ϕ_α depend only on n, r, and d.

Let now $V = V_{n,d,r}$ be a variety defined over k, not contained in a hyperplane. Let (w) be a point of P', and let $c(w) = (c_0(w), \cdots, c_M(w))$ be the Chow coordinates of the cycle $V \cdot H_w$. If $F(U^{(0)}, \cdots, U^{(r)})$ is the associated form of V, then it can be verified that the associated form of $V \cdot H_w$ is $F(w, U^{(0)}, \cdots, U^{(r-1)})$. Hence each $c_i(w)$ is a form of degree d in the quantities (w), with coefficients in k.

Let R be the algebraic set of points (w) in P' such that $V \cdot H_w$ is not a variety. This algebraic set R is defined over k, and we have $(w) \in R$ if and only if $c(w) \in C$, i. e. if and only if $\phi_\alpha(c(w)) = \phi_\alpha(c_0(w), \cdots, c_M(w)) = 0$ for all α. If $(W) = (W_0, \cdots, W_n)$ are the variables of P', at least one of the polynomials $\phi_\alpha(c(W))$ does not vanish identically. (If they all did, R would be the entire space P', which is not true.) The degree of this $\phi_\alpha(c(W))$ depends only on the degree of ϕ_α and on the degree d of each $c_i(W)$. Since the ϕ_α depend only on n, d, and r we have proved that R is contained in a hypersurface $\phi_\alpha(c(W)) = 0$ with coefficients in k, whose degree is a constant e depending only on n, d, r. This hypersurface defines a positive cycle Z in P^n, of degree e and dimension $n-1$, rational over k. By Lemma 1, we have $N_R \leqq N_Z \leqq A_1(n, e, n-1) q^{n-1}$ where A_1 is the constant of Lemma 1. Putting $A_2(n, d, r) = A_1(n, e, n-1)$, we see that Lemma 2 is proved.

Remark. All the constants depending on n, d, r which we are finding in this paper can easily be estimated, except for e in the proof of the preceding lemma.

We are now in a position to prove the main theorem.

Let $V = V_{n,d,r}$ be defined over k. We may assume that V is not contained in a hyperplane. Indeed, if V is contained in a hyperplane, or in a linear variety, then it is easily verified that the smallest linear variety L_0 containing V is also defined over k. (The linear forms defining L_0 are precisely the linear forms contained in the prime homogeneous ideal in $k[X]$ defining V.) L_0 can then be viewed as a projective space of dimension $n_0 \leqq n$, and it would suffice to prove our theorem for V as a subvariety of L_0.

If V is not contained in a hyperplane, then the cycle $V \cdot H$ is defined for every hyperplane H, and Lemma 2 will be applicable.

We consider the pairs $((x), H)$ consisting of a point (x) of V in k and a hyperplane H in P'^n defined by an equation $\sum w_i X_i = 0$ with $w_i \in k$ and passing through (x), i. e. such that $H(x) = \sum w_i x_i = 0$. We shall count these pairs in two ways.

We shall denote by κ_{n+1} the number of points in k in the projective space P^n. This is given by $\kappa_{n+1} = q^{n+1} - 1/q - 1 = 1 + q + \cdots + q^n$. This is also the number of hyperplanes H of P' in k. Similarly, the number of hyperplanes in k passing through a given point (x) in k is κ_n. From these remarks, it follows that the number of pairs $((x), H)$ is given by the following two equal numbers:

$$N\kappa_n = \sum_H N_{V \cdot H}$$

where N is the number of points of V in k, $N_{V \cdot H}$ is the number of points of the cycle $V \cdot H$ in k, and the sum is taken over all hyperplanes H in k. Solving for N, and using the same notation as in Lemma 2, we see that N is equal to

$$\frac{1}{\kappa_n} \sum_{H \notin R} N_{V \cdot H} + \frac{1}{\kappa_n} \sum_{H \in R} N_{V \cdot H} \tag{1}$$

where R is the algebraic set of hyperplanes H such that $V \cdot H$ is not a variety.

If N_R is the number of terms in the second sum, i. e. the number of hyperplanes H in R, the number of terms in the first sum is $\kappa_{n+1} - N_R$. We shall now estimate the two sums, and begin with the first one.

For $H \notin R$, $V \cdot H$ is a variety of degree d, dimension $r - 1$, defined over k. It follows from the induction hypothesis that for $H \notin R$, we have $| N_{V \cdot H} - q^{r-1} | \leqq \delta q^{r-3/2} + A_3 q^{r-2}$ where $A_3 = A(n, d, r-1)$ is the constant determined inductively, depending only on n, d, and $r - 1$. Summing over $H \notin R$, and dividing by κ_n we get

$$\left| \frac{1}{\kappa_n} \sum_{H \notin R} N_{V \cdot H} - Q \right| \leqq \frac{\kappa_{n+1} - N_R}{\kappa_n} (\delta q^{r-3/2} + A_3 q^{r-2})$$

where $Q = \kappa_n^{-1}(\kappa_n - N_R)q^{r-1}$. We note that $\kappa_{n+1} = q\kappa_n + 1$, and $q^{n-1} \leqq \kappa_n$. Also we know by Lemma 2 that $N_R \leqq A_2 q^{n-1} \leqq A_2\kappa_n$. This shows that $| Q - q^r | \leqq (1 + A_2)q^{r-1}$ and that $\kappa_n^{-1}(\kappa_{n+1} - N_R) \leqq \kappa_n^{-1}\kappa_{n+1} \leqq q + 1$. From this we conclude immediately that

$$(2) \qquad \left| \frac{1}{\kappa_n} \sum_{H \notin R} N_{V \cdot H} - q^r \right| \leqq \delta q^{r-\frac{1}{2}} + A_4 q^{r-1}$$

with a suitable A_4.

We now turn to our second sum, $\sum_{H \epsilon R} N_{V \cdot H}$, and we shall prove that it can be absorbed by the error term involving q^{r-1} only.

If $H \epsilon R$, then $V \cdot H$ is a cycle of degree d, dimension $r - 1$, and rational over k. By Lemma 1, it follows that $N_{V \cdot H} \leqq A_1 q^{r-1}$ for a suitable constant A_1. By Lemma 2, we know that $N_R \leqq A_2 q^{n-1}$. Hence

$$(3) \qquad \frac{1}{\kappa_n} \sum_{H \epsilon R} N_{V \cdot H} \leqq \frac{q^{n-1}}{\kappa_n} A_1 A_2 q^{r-1} \leqq A_5 q^{r-1}$$

because $q^{n-1} \leqq \kappa_n$, as we have already remarked.

Combining (1), (2), and (3) we see that there exists a constant A_6 (the desired constant $A(n, d, r)$) such that $| N - q^r | \leqq \delta q^{r-\frac{1}{2}} + A_6 q^{r-1}$. This concludes the proof of the theorem.

2. Corollaries and applications. We list some of the immediate corollaries to Theorem 1.

In the first place, our theorem is of the nature of an asymptotic result. If k_ν is the extension of degree ν over k, and $N^{(\nu)}$ is the number of points of V in k_ν then

$$N^{(\nu)} = q^{\nu r} + O(q^{\nu(r-\frac{1}{2})}) \quad \text{for} \quad \nu \to \infty.$$

In particular, if $\nu \to \infty$ then $N^{(\nu)} \to \infty$ also. We shall now see that this asymptotic behavior can be stated more generally for abstract varieties.

Let $V = V_{n,d,r}$ be a variety defined over k, and let F be a frontier on V (i. e. an algebraic set properly contained in V) also defined over k. Let $N^{(\nu)}{}_{V-F} = N^{(\nu)}{}_V - N^{(\nu)}{}_F$ be the number of points of $V - F$ which are in k_ν. Then from Lemma 1 we see immediately that $| N^{(\nu)}{}_{V-F} - N^{(\nu)}{}_V | \leqq Bq^{\nu(r-1)}$ where B is a constant depending on V and on F.

If V and V' are two varieties of dimension r in P^n, defined over k, and if T is a birational correspondence between them, also defined over k, then there exist frontiers F and F' defined over k such that T is everywhere biregular in $V - F$ and $V' - F'$. T gives a $1 - 1$ correspondence between the points of $V - F$ and those of $V' - F'$, so $N^{(\nu)}{}_{V-F} = N^{(\nu)}{}_{V'-F'}$.

Let now $V=(V_\alpha, F_\alpha, T_{\beta\alpha})$ be an abstract variety, represented by varieties V_α in projective space, with frontiers F_α, and birational correspondences $T_{\beta\alpha}$, all of which are defined over k. If P is a point of V, then P has representatives P_α in some of the V_α. If one of these P_α is rational over k, then so are all the other representatives, and hence in this case we may say that P is rational over k.

If K is a function field of dimension r over k, then we shall say that an abstract variety V is a model of K if V is defined over k, and if $K=k(P)$ where P is a generic point of V over k. The following two corollaries of Theorem 1 are immediate consequences of the preceding discussion.

COROLLARY 1. *Let K be a function field of dimension r over a finite constant field k. If V is any abstract model of K, and $N^{(\nu)}$ the number of points of V in k_ν, then $N^{(\nu)}-q^{\nu r}$,* mod $O(q^{\nu(r-1)})$, *is a birational invariant (i. e. does not depend on the choice of the model V of K).*

COROLLARY 2. *Let K/k be as in Corollary 1. There exists a constant γ such that for any abstract model V of K, we have*

$$|N^{(\nu)}-q^{\nu r}|\leqq \gamma q^{\nu(r-\frac{1}{2})}+Bq^{\nu(r-1)}$$

where B is a constant depending on V. If K has a projective model of degree d, then we can take $\gamma\leqq(d-1)(d-2)$.

The smallest constant γ that can be selected in Corollary 2 is obviously a birational invariant, and we shall return to discuss it below. For the moment, we note that an abstract model of the function field K can be projective or affine, and can also be chosen without singularities, because the singular points on a variety defined over k are a frontier F, properly contained in V and defined over k. Hence our corollaries show that as ν increases indefinitely, the number of simple points on V in k_ν also increases indefinitely.

COROLLARY 3. *Let V be an abstract variety defined over k. Let m be a given integer. There exists a zero dimensional positive cycle on V, rational over k, and of degree prime to m. Hence there exist cycles on V which are zero dimensional, rational over k, and of degree* 1.

Proof. From the above remark and the corollaries, there exist non singular points of V in the field k_ν when ν is sufficiently large and prime to m.

The preceding result generalizes the well known theorem that a function field in one variable over a finite constant field always has a divisor of degree 1. In this vein, we can state the following invariant result:

COROLLARY 4. *Let K be a function field over a finite constant field k. Let $x_1, \cdots, x_n$ be a finite set of non zero elements of K. For all ν sufficiently large, there exists a place ϕ of K which is k_ν-valued, and such that $\phi(x_i) \neq 0$, $\neq \infty$ for any i.*

Proof. After enlarging our set (x) by a finite number of elements if necessary, we may assume that $k(x) = K$, and that for each i, x_i^{-1} appears in the set (x). Under these assumptions it suffices to prove that there exists a place ϕ of K which is k_ν-valued and such that $\phi(x_i)$ is finite for all x_i. We view (x) as the generic point of an affine variety V defined over k. By Corollary 1, for all sufficiently large ν, there exists a point (x') of V in k_ν which is simple on V. If $\mathfrak{o}$ is the specialization ring of (x') in K and $\mathfrak{p}$ the maximal prime ideal, then $\mathfrak{o}/\mathfrak{p}$ is isomorphic to the subfield $k' = k(x')$ of k_ν. (The isomorphism is induced by the map $f(x) \to f(x')$, $f \in k[X]$). Since k is perfect, the completion of $\mathfrak{o}$ is isomorphic to a power series ring $k'\{t\} = k'\{t_1, \cdots, t_r\}$ and K can be identified with a subfield of the quotient field Ω of $k'\{t\}$. It is clear that there exists a k'-valued place ϕ of Ω mapping each t_j on 0. (For instance, view Ω as a subfield of the repeated power series field $\Gamma = k'\{t_1\}\{t_2\} \cdots \{t_r\}$. By mapping successively each t_j $(j = r, \cdots, 1)$ on 0, we get a k'-valued place ϕ of Γ.) Since the quantities x_i are in $\mathfrak{o}$ for all i, the place ϕ is finite on the x_i. The restriction of ϕ to K gives the desired place.

We now turn to the applications of Corollaries 1 and 2 to the zeta function. Let V be an abstract variety of dimension r, defined over k. We associate with V the analytic function $Z(U)$ defined by

$$d \log Z(U)/dU = \sum_{\nu=1}^{\infty} N^{(\nu)} U^{\nu-1}.$$

If V is complete and without multiple points, then $Z(U)$ is the zeta function of V [5] and Weil's conjectures state among other things that $Z(U)$ is a rational function, satisfying a functional equation of the usual type. If V has singular points, or is not complete, then the function $Z(U)$ is probably still a rational function, but it need not satisfy a functional equation.

We shall not at this point go into a detailed analysis of the relation between the functions $Z(U)$ associated with arbitrary varieties, and those associated with complete varieties without multiple points, because the results which we shall now prove as a consequence of Corollary 2 will be applicable to any variety.

COROLLARY 5. *Let V be an abstract variety of dimension r, defined over k. Then the associated function $Z(U)$ has no pole or zero in the circle*

$|U| < q^{-r}$, and has exactly one pole of order 1 in the circle $|U| < q^{-(r-\frac{1}{2})}$, namely at $U = q^{-r}$.

Proof. According to Corollary 2, we can write $N^{(\nu)} = q^{\nu r} + A_\nu q^{\nu(r-\frac{1}{2})}$, where A_ν is a constant, bounded in absolute value by a fixed constant A. This gives

$$d \log Z(U)/dU = q^r \sum_{\nu=0}^{\infty} (q^r U)^\nu + \sum_{\nu=1}^{\infty} A_\nu (q^{r-\frac{1}{2}} U)^\nu.$$

The first of these series is the geometric series and defines the function $1/(1-q^r U)$. Since $-q^r/(1-q^r U) = d \log (1-q^r U)/dU$ we see that $d \log [Z(U)(1-q^r U)]/dU = \sum_{\nu=1}^{\infty} A_\nu (q^{r-\frac{1}{2}} U)^\nu$. Hence $Z(U)(1-q^r U)$ has no pole or zero in the circle $|U| < q^{-(r-\frac{1}{2})}$ because the latter series converges in that circle. Furthermore q^{-r} is a pole, and is the only pole inside the circle, as was to be shown.

COROLLARY 6. *Let V and V' be two varieties which are birationally equivalent, and let $Z(U)$ and $Z'(U)$ be the associated functions. Then the function $Z(U)/Z'(U)$ has no zero or pole in the circle $|U| < q^{-(r-1)}$. Hence the zeros and poles of $Z(U)$ in this circle are birational invariants.*

Proof. It is clear from Corollaries 1 and 2 that the series for

$$d \log Z(U)/dU - d \log Z'(U)/dU = d \log (Z(U)/Z'(U))/dU$$

converges in the circle $|U| < q^{-(r-1)}$, and hence that $Z(U)/Z'(U)$ has no zero or pole in the circle.

Concerning the behavior of $Z(U)$ for $|U| \geqq q^{-(r-\frac{1}{2})}$ we can only make the following conjectural statements, which complement the conjectures of Weil [5].

Let P be the Picard variety of V. P is an abelian variety, whose dimension g is the irregularity of V. Let ι be the endomorphism induced on P by the automorphism $(x \rightarrow x^q)$ of the universal domain, and let

$$F(U) = \prod_{i=1}^{2g} (1 - \alpha_i U)$$

be the characteristic polynomial of ι as defined in [3], § 67. Then the series for

$$d \log Z(U)/dU + d \log (1 - q^r U)/dU - d \log F(U)/dU$$

converges in the circle $|U| < q^{-(r-1)}$, and the function $Z(U)(1-q^r U)/F(U)$ has no zero and no pole inside this circle, and has at least one pole on the circle $|U| = q^{-(r-1)}$, provided of course that V is complete and without multiple points.

Furthermore, if K/k is a function field as in Corollary 2, and if γ is the smallest constant for which Corollary 2 holds, then $\gamma = 2g$. (This gives the explicit determination of γ as a birational invariant.)

If V is a curve, then $P = J$ is the Jacobian variety, and g is the genus of the curve. For this case, all the preceding statements are well known and are contained in [2], except for the last, i. e. that $\gamma = 2g$ is actually the best possible constant for the inequality $|N^{(\nu)} - q^{\nu}| \leqq \gamma q^{\nu/2} + B$ of Corollary 2. As to this, it obviously suffices to prove it in the case that V is complete and without multiple points, and then it amounts to proving that $\gamma = 2g$ is the best possible constant for the inequality $|1 + q^{\nu} - N^{(\nu)}| \leqq \gamma q^{\nu/2}$. Actually, this last statement is implicit in [2], as the following argument shows.

We have ([2], § 22, p. 71) $|1 + q^{\nu} - N^{(\nu)}| = \sigma(\iota^{\nu}) = \sum_{i=1}^{2g} \alpha_i^{\nu}$ and $|\alpha_i| = q^{\frac{1}{2}}$, so that we can write $\alpha_i = q^{\frac{1}{2}} \exp(2\pi i \theta_i)$ where θ_i is a real number between 0 and 1. Hence we can write

$$\sigma(\iota^{\nu}) = q^{\frac{1}{2}} \sum_{i=1}^{2g} \exp(2\pi i \nu \theta_i).$$

For infinitely many ν, $\sum_{i=1}^{2g} \exp(2\pi i \nu \theta_i)$ will be as close to $2g$ as we please, and hence $2g$ is indeed the best possible constant.

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